Transforming care for people with peripheral artery disease through community interventions

Please describe your research career to date (1116/1000)

I have a ***strong track record of high-quality interdisciplinary research*** in the rehabilitation of long-term condition that impacts beneficially on clinical practice and patient outcomes. My research on consequences and rehabilitation of inflammatory conditions and has now broadened to include other painful long-term conditions (e.g., peripheral artery disease (PAD)). My total grant income is £11.8million (£8.8million since 2021), with £2.9million funding pending.

My research is at the forefront of the national agenda to shift care into the community. I led the team that identified the factors that influenced home-based walking exercise and tested the first, home-based walking exercise behaviour change intervention for PAD patients (MOSAIC1). This research was published in JAMA *2022,* showcased in a JAMA editorial and broadcast by multiple mass media organisations. It informed international guidance[1] The feasibility and implementability of MOSAIC1 are now being investigated in Scotland[2] and informed an NIHR PGfAR application (Role: Chief Investigator, stage 2, pending).

I led one of the first research programmes to identify poor sensorimotor function, functional performance, and quality of life in adults with rheumatoid arthritis (RA). I provided key evidence on the safety and effectiveness of exercise for reversing these debilitating effects, improving patient quality of life and participation. The impact of my research was recognised by the British Society of Rheumatology (BSR Prize 2002) and Chartered Society of Physiotherapy (Robert Williams Silver medal 2003). I designed clinically applicable, novel interventions and led trials that showed the effectiveness and, cost effectiveness of a RA upper limb exercise programme and early RA gait rehabilitation[3]. My research informed guidelines and drove the integration of exercise into the management of RA. It informed

- NICE guidelines (NG100)
- BSR guidelines for the management of pain in Inflammatory Arthritis (2022)
- BSR guidelines for management of Foot Health in Inflammatory Arthritis (2024)

The excellence and impact of my research was recognised when I was invited to give a BSR 'Jewels in Crown' keynote (2015), received the European Alliance of

Associations for Rheumatology (EULAR) research excellence award (2019) and co-author the 'Oxford Textbook of Rheumatoid Arthritis (2022).

My ***collaborations*** with European experts from over 44 countries have enabled participation in some of the most impactful European guideline development group[4]. I am lead author on international outputs that drive practice change including researchers from USA, Sweden, Australia, New Zealand and Denmark[5-7]. These partnerships have sparked research collaborations (NIHR150711) and invitations to act as international PhD supervisor[8,9]. I lead partnerships that are pushing forward the transition of care to community rehabilitation. I designed and secured funding for a consortium of over 32 researchers, professional and voluntary organisations to deliver the NIHR-supported Community Rehabilitation Research Incubator. This enabled participation in the Community Rehabilitation Alliance, a coalition of over 50 charities and professional bodies, driving the pace of change in commissioning, planning and delivery of evidence-based rehabilitation nationally. I am invited to work in partnership with community and acute clinical service planners as part of our Integrated Care Board research working group to develop research that addresses local patient needs and priorities.

My ***research leadership*** is defined by my role as an academic leader in rehabilitation research and as a pioneer of physiotherapy research.

- I am a panel member for NIHR Pre-Clinical Academic and practitioner Fellowships, Wellcome Trust Global Health Doctoral Fellowships, and former member of NIHR RfPB, acting as lead assessor.
- As Chair of the D-MAPP programme committee [NIHR201612] and TSC/DMC member for three trials, I provide strategic oversight of substantial NIHR funding awards.
- In recognition of my expertise, I regularly review international grants submitted on rehabilitation calls e.g. Dutch Physiotherapy Society.
- I was the first physiotherapist invited to be an Associate Editor for Rheumatology Advances Practice and contributed to all aspects of the editorial process including editing the inaugural special issue on 'Exercise as medicine: Physical Activity in Inflammatory Arthritis'.
- As Trustee and Chair of the British Health Professions in Rheumatology scientific committee and member of the BSR conference committee I led the research prioritisation, reviewed proposals, curated and showcased high-quality research at the annual conference.

I lead flagship ***translational research capacity building*** activities. As a NIHR Knowledge mobilisation Fellow, my team develop and deliver the national NIHR

strategy for maximising research impact. I shaped the new NIHR application form/guidance, developed researcher knowledge mobilisation capability through a national webinar series and designed and delivered 'maximising impact' training for NIHR RSS advisors. I will lead national translational researcher capability as a founding member of the NIHR-supported Knowledge mobilisation Incubator.

My substantial ***track record in developing researchers*** exemplifies my interdisciplinary rehabilitation research programme. I developed and secured funding for the NIHR-supported Community Rehabilitation Research Incubator that builds researcher capability and capacity in multiprofessional community practitioners. As physiotherapy lead for the HEE Integrated Clinical Academic and NIHR Insights South-East London programme I promote research opportunities to pre-registration students to inspire the next generation of allied health professional researchers.

I effectively coach research students through fellowship applications, researcher development, PhD examinations and postdoctoral awards. I've supervised six PhD students to completion on time. I currently supervise three PhD students and three clinicians completing NIHR predoctoral awards.

Through the BSR and NIHR mentoring programme, I mentor multidisciplinary researchers at all levels. I recently coached one physiotherapy clinician to successfully secure her first externally funded postdoctoral award. I mentored a postdoctoral health psychologist to obtain her first tenured academic position. I am now coaching her to supervise her first PhD student and develop her advanced fellowship application. I mentor a medically qualified data scientist and exercise physiologist NIHR Advanced fellows.

I am passionate about ***inclusive patient involvement*** and develop innovative strategies to maximise involvement from underserved communities. I established the *City St George's vascular public advisory group.* By partnering with patient and community groups (e.g., national PAD advisory group, Eastern European and Romanian hub), our group now involves geographically diverse patients from PAD underserved communities. Patients drive the identification of important questions and partner us to deliver and disseminate impactful research using creative strategies. To enhance involvement, I provide technology support, plain and multiple language summaries, research training and reimburse expenses.

This NIHR Professorship will be a step change into leading multiple research projects of international importance and a strategic leadership role aligned with

the NIHR aims and mission for high-quality community rehabilitation research. This professorship will provide me with the support I need to develop my long-term vision: to radically shift the scientific understanding, quality of care and equity of access to community interventions for PAD patients. The financial and institutional commitment offered by the scheme and City St George's will enable me to build my research group, increase research activity, support community rehabilitation research capacity building and maximise impact.

Contextual factors (480/500 words)

Delayed commencement of research career. As a clinical physiotherapist I had limited opportunities to experience or contribute to research. I did not publish any research. As rehabilitation research was in its infancy and physiotherapy research roles/mentorships were limited, I was unaware of how take the first steps into research. Despite this, I self-funded an MSc Sports Science at Loughborough University and, after returning to full time clinical practice for 5 years (with no dedicated research time), I obtained a small Arthritis Research Campaign bursary (£6,000) to fund PhD fees.

Lack of postdoctoral research opportunities; I completed my PhD viva when pregnant with my first child. Despite substantial efforts to identify suitable research roles, I returned to clinical practice after parental leave, and did not complete any dedicated post-doctoral research. I commenced an educational position in a university teaching department with minimal protected research time. This limited the development of research collaborations, opportunities to (co) write research grants and publish.

Maternity leave and caring responsibilities. I have had two periods of maternity leave (one year each). One immediately after I completed my PhD and another after I had completed my first year as a (teaching) lecturer.

Part-time working and high educational/administration load. Due to caring responsibilities, I worked as a part-time teaching lecturer (0.5 WTE,2002-2013), then Senior Lecturer (0.6 WTE 2013-2021). The primary focus of my department was delivering three teaching intensive allied health professional programmes. Educational activities accounted for at least 60% of my part-time roles. I delivered a substantial teaching load, led three teaching modules, chaired the postgraduate physiotherapy programme board and was postgraduate research co-ordinator/ Chair of the postgraduate committee. For one year, I was Programme Director for the MSc in Advanced Physiotherapy (0.8 WTE, 80% education). I had little research time or support, despite working in a research intensive university.

To increase research time and support, I changed intuitions in 2021. City St George's provided more research time, development opportunities, leadership training and encouraged me to apply for this award. I have increased outputs, postgraduate students, research networks and funding. Whilst my teaching load is minimised, I am seconded part-time to NIHR as a Senior Research Fellow (Knowledge Mobilisation) so my research time remains restricted (0.4 WTE).

Together, these circumstances have substantially impacted my research trajectory including number of outputs, PhD students, research income, collaborations and the delivery of my ambitious research programme. This award will provide be a step change for my career. It will release me from teaching and NIHR secondment which inhibit my research and leadership capability and enable me to accelerate my team building. The beneficiaries will be patients, clinicians, the NHS and academic community. This application has a secure foundation, with guaranteed deliverables, alongside explorative transformative projects.

Plain English Summary (595/600 words)

Why is this research needed?

Community rehabilitation improves people's health and wellbeing and saves money by treating people at home. But community rehabilitation is not provided equally, and it is not known how to best to treat people with some conditions or whether treatments are acceptable to patients in the community. Improving equal access to effective and acceptable community rehabilitation is central to the new government's NHS plan.

One of the conditions that we don't know how to rehabilitate in the community is called peripheral artery disease. Peripheral artery disease is poor leg circulation due to narrowed arteries. People from Black, South Asian or Eastern European communities, those with diabetes or living in areas with socioeconomic deprivation are affected more by peripheral artery disease.

Severe peripheral artery disease can cause constant leg pain, sores and gangrene in the toes. This is called chronic limb threatening ischaemia. Most people need their arteries unblocked in hospital and then need to self-manage their recovery at home. This is difficult but there are no self-management programmes to support people with chronic limb threatening ischaemia to recover and live well at home.

Some people with milder peripheral arterial disease experience leg pain on walking (called intermittent claudication). Walking frequently is the best way to improve this pain and reduce the risk of severe artery disease. It is hard for people to walk on their own when they are in pain. New ways are needed to help people walk regularly, improve their leg pain and do more activities.

Most people with peripheral artery disease are not active enough to reap the health benefits. This increases their risk of getting other health problems. There are many ways to help people to become active. One way is to use a smartwatch that records how much activity is done and sends messages to encourage people to be more active. Smartwatches can be used separately or as part of an activity programme. But some older people find smartwatches difficult to understand and use.

Aims

To work with patients, practitioners and service planners to co-design ways to provide community rehabilitation for everyone with peripheral artery disease and explore new ways to help people to be more active. People who are most affected by peripheral artery disease will be included.

What will happen?

I will build a team to

(i) summarise the evidence, interview patients and practitioners and co-design a self-management programme to help people with chronic limb threatening ischaemia live well at home.
(ii) look at the results from a study of an existing walking exercise programme for people with intermittent claudication and use this information to adapt this programme so it can be provided in the community.
(iii) interview patients and practitioners, co-design and user test a new acceptable and usable treatment includes a smartwatch to help everyone with peripheral artery disease be more active.
(iv) co-design a plan to implement these treatments in the NHS, if these ways are shown to be effective and acceptable.

How have patients and communities been involved?

Members of the public and patients from different backgrounds chose this topic and helped design this research. They helped write this summary. The public advisory group will meet every 6-months and be involved in workstream management.

Sharing this research

Patients, practitioners, service planners and voluntary and professionals' organisations (like Circulation Foundation) will plan and help share our research progress and findings, so it is available in the NHS and for the public. This will include writing scientific reports; summaries for clinical guideline organisations and service planners; giving talks; hosting meetings; designing infographics, animations, websites and social media.

Scientific summary (500/500)

Provision of community rehabilitation is inequitable. Many interventions are not evidence-based or acceptable. Rehabilitation of some high burden conditions, such as peripheral artery disease (PAD), receive relatively little national support, prioritisation or research funding, fuelling health inequalities.

PAD affects over 237 million adults globally. One PAD symptom is intermittent claudication (IC) although in its most severe manifestation, PAD leads to chronic limb threatening ischaemia (CLTI). Over half of those with symptomatic PAD will die, have an amputation or cardiovascular event within five years of diagnosis. CTLI prognosis is extremely poor; commensurate with terminal cancer. The economic burden of PAD is huge and rising.

PAD patients report dissatisfaction with care and improving outcomes in PAD patients across the symptom range is the highest ranking research priority among all vascular conditions. Providing PAD rehabilitation and supporting self-management will save money. Yet, equitable access to community rehabilitation is compromised by lack of acceptable, feasible, interventions that are not designed with patients or community practitioners, restricted resources and a high and increasing demand from a diverse patient group with complex needs. Many people with PAD are also excluded from research and unable to access or use new treatments e.g. digital interventions.

New scalable, co-designed, inclusive PAD interventions such as practitioner supported community rehabilitation, self-management programmes and 'next-generation' mobile-based health technologies (mhealth) are promising. Yet, evidence is mixed, none have been designed with PAD patients, or commissioned.

The aim of this professorship is to transform the care for people with peripheral artery disease through community interventions. Over five work packages, I will lead research that will:

1. Maximise diverse and inclusive involvement, participation and engagement of underserved groups with PAD

2a Through systematic review, qualitative and participatory approaches, codesign an inclusive supported self-management intervention to improve quality of life in CLTI patients. 2b. co-design a funding application to investigate inclusivity, effectiveness and cost effectiveness

3a. Through secondary analysis of existing quantitative and qualitative data, participatory approaches, co-refine and feasibility test an inclusive home-based walking exercise behaviour-change programme for people with IC for delivery by multidisciplinary healthcare and non-traditional healthcare exercise providers in primary care and community services. 3b evaluate it in a trial

4a Through qualitative approaches, co-design, refine and user test an inclusive mhealth intervention for PAD patients to increase physical activity 4b. design a study to test usability, inclusivity, clinical and cost-effectiveness

5. Drawing on data from workstreams 2-4 specify the implementation context, outcomes and mechanisms and co-design tailored implementation strategies for inclusive community PAD rehabilitation services

This ambitious programme of work strikes a balance between building on my existing work, with guaranteed high impact outputs, and projects which promise a step change in trajectory within five years. It will spearhead the shift of PAD care into the community and develop research capacity in under-represented disciplines.

Mitigating research and digital exclusion and involving patients and the public are central to this research. Patients identified the research questions, helped plan this research and will help deliver and mobilise research progress and findings using co-designed, systematic, creative approaches.

Background (5447/5000words)

1. The problem

Community rehabilitation is person-centred, out-of-hospital, care tailored for people with different conditions and fluctuating needs. Globally, one in three people require rehabilitation and ageing populations and increasing numbers of people living with long-term conditions are fuelling demand[10]. In 2017, the World Health Organisation recognised that many countries, including the UK, are not providing appropriate and equitable rehabilitation nor are they equipped to address future rehabilitation needs. Their call to action urged countries to scale-up evidence-based community rehabilitation to address health and social inequities as a critical health priority[10].

In the UK, provision of community rehabilitation is inconsistent and many interventions are not evidence-based, feasible or acceptable to patients[11]. Rehabilitation of some high burden conditions, for example peripheral artery disease (PAD), have received relatively little national support, prioritisation or targeted research funding to date compared to other cardiovascular diseases (e.g., coronary heart disease or stroke), fuelling health inequalities[12].

Peripheral artery disease affects over 237 million adults globally and 15-20% of the UK population aged over 55 years[13,14]. It frequently occurs alongside

other conditions e.g., diabetes, high blood pressure or cholesterol[15,16] and disproportionally affects socioeconomically deprived populations, and South Asian, African Caribbean, and Eastern European groups[15,17-19]. It causes substantial disability, increased cardiovascular morbidity and mortality and is associated with feelings of powerlessness, isolation and poor quality of life[14,16,20,21]. Its economic burden is huge and cardiovascular-related healthcare costs alone in England for people with PAD are £1.4 billion/year, with wider economic annual costs approximately £4 billion[22,23].

One PAD symptom is walking leg pain (intermittent claudication, IC) although in its most severe manifestation, PAD leads to rest pain, and tissue loss (chronic limb threatening ischaemia (CLTI))[24]. Despite optimal treatment, over half of those with symptomatic PAD will die, have an amputation or cardiovascular event within five years of diagnosis[15,16,20,25,26]. CTLI prognosis is particularly poor, even after revascularisation; commensurate with terminal cancer[27].

Improving PAD outcomes is the highest ranking James Lind Alliance research priority among all vascular conditions[28].

PAD care is delivered through a vascular network of secondary care hub hospitals supported by community and primary care spoke services. Guidelines recommend medical optimisation, lifestyle modification, revascularisation/limb salvage, community rehabilitation and home-based self-management support (including personalised home-based exercise, digital interventions)[29,30,31]. Yet, equitable access to personalised community rehabilitation that develops an individual's capability, opportunity and motivation to avoid deterioration in and self-manage their symptoms and condition at home is compromised by lack of acceptable evidence-based interventions and a high and increasing demand from a diverse patient group with complex needs[11,32]. Many interventions are not designed with patients or community practitioners and are not feasible for practitioners to deliver due to restricted knowledge, skills and resources (technology, time, space, training, staff)[11].

Community rehabilitation for PAD is an area of historical under investment and high research need.

Equity factors influence access to healthcare, health opportunities and outcomes, but less than a quarter of people with PAD are screened for eligibility to participate in rehabilitation trials and PAD patients are typically excluded based on at least one equity factor[33,34]. Yet, these subgroups may respond

differently to community rehabilitation due to differences related to social, environmental, physiological or disease states. So, evidence when available, doesn't always relate to those most in need. This is important because findings from PAD trials and systematic reviews of trials inform practice guidelines and service planning so excluding these subgroups may mean that community interventions may not be as effective once implemented.

These gaps need urgently addressing, and subgroups with higher prevalence or disease burden, or those who do not receive or cannot access similar levels of services need to be included in PAD intervention development and research so evidence-based community interventions can be implemented equitability and at scale.

2. Importance

PAD patients report dissatisfaction with care[35]. Low engagement (<50% enrolment) and high attrition (30% non-completion) in hospital-based PAD rehabilitation is driven by social (e.g. family support), physical (e.g. pain), and environmental factors (e.g. travel)[36,37,38]. Women, older adults, CLTI patients, people from low socioeconomic or ethnic communities are disproportionately affected by these barriers[39,40]. New scalable, co-designed, inclusive interventions are needed such as practitioner supported community rehabilitation, self-management programmes and mobile-based health technologies (mhealth)[41,42]. These approaches will save money as patients who feel most confident in managing their longterm condition have 38% fewer emergency admissions and 32% fewer emergency hospital attendances than those who feel least confident[43].

Public advisors and practitioners support this research because it addresses avoidable deterioration in PAD and health from inequitable NHS access. It will widen access to community care and address key goals set out in the 2015 WHO Global Goals pledge (e.g., ensuring equitable access to community interventions)[32].

This research focusses on five James Lind Alliance PAD priorities; How can we a) improve patient outcomes b) make it easier for patients to get help for this problem? c) help better educate patients with poor leg circulation? d) improve exercise provision and access? and e) what are the best ways to reduce leg pain without an operation?[28]
It aligns with National Institute for Health and Care Excellence (NICE) [CG147] research recommendation (optimal exercise therapy for PAD)[29] and NIHR strategic themes[44].

Crucially, it is congruent with the Department of Health and Social Care areas of research interests (early action to prevent poor health outcomes, accelerating the adoption and scale-up of innovation)[45] , NHS digital health strategy[46] and recent Darzi recommendations: shift care to community settings; expand digital services; promote prevention and minimise disease progression[47].

3. Existing evidence

Comprehensive PAD rehabilitation requires a suite of evidence-based community interventions considering the severity, complexity and biopsychosocial impact of PAD and associated multiple comorbidities on everyday life.

People with CLTI have severely restricted lifespan, high symptom severity and co-morbidities resulting in poor quality of life and high social isolation, even after revascularisation[27]. Systematic reviews support the effectiveness of short self-management approaches in people with long-term and life limiting conditions on quality of life and independence[48,49,50]. When people receive personalised support sessions their emotional wellbeing improves because they feel more in control of their symptoms and condition, have greater confidence, and gain more knowledge and strategies to manage everyday life[48-50]. Evidence for self-management interventions also emphasised the need for contextualisation to 1) specific challenges and complexity of CLTI and personal and cultural characteristics 2) understanding the community setting and 3) training which addresses practitioners' knowledge, skills and attitudes[51,52]. No community supported self-management interventions have been co-designed with CLTI patients or commissioned in the NHS.

Guidelines recommend supervised exercise programmes as first-line treatment for IC as evidence is insufficient to recommend home-based walking exercise, given inconsistent trial results[29,53]. Systematic reviews concluded that home-based interventions are inferior to hospital based exercise but better than providing no or basic exercise advice[54,55]. Most studies were small, and outcomes were very low or low quality. Heterogeneous intervention delivery methods, settings, intervention components that do not include tailored behaviour change techniques, exercise dose variability, impractical interventions, inappropriate outcomes and/or poor treatment fidelity all contribute to uncertainty about whether home-based exercise interventions are effective.

MOSAIC1, a feasible and acceptable home-based walking exercise behaviour change intervention, addresses these limitations (Figure 1)[56-59]. In an efficacy trial, participants with IC increased mean 6-minute walk distance by a clinically

meaningful difference, compared with usual care[57]. MOSAIC1 needs adapting for delivery by community providers to diverse patients across the UK. This informs my NIHR PGfAR application (Stage 2).

Physical activity is a tertiary prevention in PAD that reduces the need for revascularisation, and reduces post-revascularisation complications in those requiring surgery, as well as the need for further complex and costly interventions. Yet, less 50% PAD patients meet physical activity guidelines for general cardiovascular health[60,61] or for improving IC[62].

The COVID-19 pandemic prompted community services to incorporate mhealth to deliver care and offer preventative strategies such as promoting physical activity. Yet, older people often experience digital exclusion limiting engagement[42]. Whilst mhealth interventions increase physical activity in the general population, the effects in PAD patients is unclear[63]. A Cochrane review comparing mhealth interventions versus usual care, exercise advice or supervised exercise included only four trials and found no clear evidence of an effect of mhealth interventions on walking or other health outcomes in people with IC[41]. All included studies had unclear or high risk of bias, excluded CLTI patients or people after revascularisation, and two studies enrolled majority white participants, limiting generalisability[41].

Mhealth interventions are evolving and now combine consumer activity trackers (i.e. Smartwatches) with 'just in time adaptive interventions' (JITAI)– personalised motivational messages delivered in response to activity feedback and practitioner support[64]. Trials incorporating mhealth interventions alongside strategies to mitigate digital exclusion show promising results[65]. Whilst a PAD message library has been developed, these are untested[66]. These 'next generation' mhealth interventions potentially provide an exciting, low-cost, adaptable and scalable PAD treatment. If co-designed with PAD patients and integrated with digital inclusion strategies, they could offer either a standalone community intervention or be incorporated interventions such as MOSAIC1 to support long-term self-management.

Why is this research needed now?

My research has started to influence practice and now needs expanding to address two key underserved groups - those with CLTI and those experiencing research and digital exclusion. It addresses a crucial unmet health need; the avoidable deterioration in PAD and health from inequitable access to community rehabilitation. It will provide new community interventions and workforce training for integration into the PAD care pathway that can be adapted for different contexts and populations.

4. Aims and objectives

Aim:

To transform the care of people with PAD across the symptom range through community interventions

Objectives:

1. Maximise diverse and inclusive involvement, participation and engagement of underserved groups with PAD in all workstreams

2a Through systematic review, qualitative and participatory approaches, codesign an inclusive supported self-management intervention to improve quality of life in CLTI patients. 2b. co-design a funding application to investigate inclusivity, effectiveness and cost effectiveness

3a. Through secondary analysis of existing quantitative and qualitative data, participatory approaches, adapt and feasibility test an inclusive home-based walking exercise behaviour-change programme for people with IC for delivery by multidisciplinary healthcare and non-traditional healthcare exercise providers in primary care and community services. 3b evaluate it in a trial

4a Through qualitative approaches, co-design, refine and user test an inclusive mhealth intervention for PAD patients to increase physical activity 4b. design a study to test usability, inclusivity, clinical and cost-effectiveness

5. Drawing on data from workstreams 2-4 specify the implementation context, outcomes and mechanisms and co-design tailored implementation strategies for inclusive community PAD rehabilitation services

5. Project Plan

With public advisors, I developed a programme theory that proposes the context, activities, outputs outcomes and impact of this professorship (Figure 2).

This research is informed by key frameworks (e.g., Medical Research Council framework[67], INCLUDE frameworks[68,69], Updated Consolidated Framework for Implementation Research (CFIR)[70], mHealth development and evaluation framework[71])

Workstream 1 Research Inclusion (1-60 months)
Question: What engagement strategies are needed to reach to people with PAD, particularly those from underrepresented communities?

Aim: *To maximise inclusive involvement, participation and engagement of underserved groups with PAD*

Inclusion in community rehabilitation research is at the core of the professorship, guided the EqIA[72] and INCLUDE frameworks[68,69]. In this cross-cutting workstream will focus creative engagement activities PAD under-served communities (i.e., socioeconomically deprived, or South Asian, African Caribbean, Eastern European communities) so their unique perspectives are heard and inform all aspects of this research.

Research inclusion strategies comprise:

- Appointing a dedicated co-ordinator to lead research inclusion and public partnerships. They will focus on developing trust and understanding between researchers and our underserved groups.
- Co-designing culturally appropriate engagement strategies, interventions, resources and study documentation with diverse communities to realise our vision for excellent and equitable opportunities.
- Inclusive recruitment strategies that promote participation from underserved, marginalised groups and those not usually recruited into research. This will be achieved by developing creative and inclusive communications with underserved communities and patient organisations to share research details and support participation. Approaches will be made through our co-ordinator, so organisations are not overburdened with multiple requests.
- Rewarding and recognising involvement, participation and engagement.
- Working with geographically diverse Local Research Delivery Network (LRDN) (e.g., urban, rural, north, south, coastal) and building engagement strategies with primary care and community services that serve underserved populations.
- Reviewing our participant demographic data from each workstream and discussing how our strategies are being implemented and how challenges will be overcome with public advisors. Additional targeted efforts and strategies will be applied to increase representation, where needed.

- Offering language translation, technical support, fees for software for online meetings and offering alternative ways of engaging e.g. telephone, receiving paper-based information
- Offering cultural competency and inclusive research training to all research team members, recruiting staff and providers.
- Developing tailored knowledge mobilisation plans with different communities and promoting inclusive communications to mobilise research knowledge beyond scientific publications e., infographics, videoclips, shared across a range of platforms.
- Regular monitoring and reporting of EqIA[72] and involvement, recruitment, participation and engagement rates from underserved groups. Involvement will be reported using GRIPP-2[73]

Output: (i) Diverse and meaningful engagement from underrepresented groups in all workstreams

Workstream 2 Inclusive community-based supported self-management for people with CLTI (months 1-36)
Question: Can a supported self-management programme to improve quality of life be codesigned with CLTI patients?

2.1 *Aim: To synthesize the determinants of health self-management in people with CLTI and cardiovascular conditions*
Design: Systematic review

Methods: Databases will systematically search for quantitative studies reporting determinants of health self-management in CLTI and cardiovascular conditions. Deduplication, screening, quality appraisal and data extraction will be completed by two researchers. Data will include: study characteristics; setting; population; self-management approach; determinants of self-management behaviour. Determinants will be summarised using the Theoretical Domains Framework inform an intervention programme theory[74].

2.2 Aim: To understand the experience of CLTI and self-management approaches from patient, carers/relative and practitioner perspectives

Design: qualitative study
Participants: A purposive sample of up to 30 patients and their carers/relatives, 15 multidisciplinary practitioners.

Patient recruitment will draw on the EqIA[72], INCLUDE frameworks[68,69] and workstream 1. Diverse patient participants (i.e., location, ethnicity, disease severity, protected characteristics) and carers/relatives will be recruited. No upper ages limits will be applied and patients with cognitive impairment will be included (with a nominated supporter) if they have capacity to consent. Potential participants will be identified via vascular centres, primary and community services, social media, community organisations (e.g., National vascular public advisory group, The Romanian and Eastern European Hub) our networks (e.g., Centre for Ethnic Health Research) and established relationships with NIHR LRCN with socioeconomic, ethnic and geographical diversity (South-east and south-west London, Northeast and North Cumbria: Humberside, East Midlands).

Practitioner participants will be identified from primary and community care, professional networks and interest groups (Community Healthcare Alliance of Research Trusts, Community Rehabilitation Research Incubator, Vascular Society of Great Britain and Ireland, Society for Academic Primary Care) and social media.

Methods: Online or in-person focus groups (n=6-8 participants/group) or online/in-person/telephone narrative interviews (participant preference) will explore experiences of CLTI and receiving or delivering self-management. Common themes will be identified using reflexive thematic analysis and participants' accounts used to develop a composite film of patient experiences and priorities[75].

2.3 Aim: To co-design inclusive tailored self-management resources for anyone with CLTI and provider training

Design: Using an Experienced-Based Co-Design, I will work with Bridges Self-management to codesign a supported self-management programme and provider training resources for people with CLTI. Bridges Self-Management is a social enterprise led by Professor Fiona Jones (mentor) and has built a (15 year) programme of research in personalised self-management support and has been evaluated in multiple areas of healthcare[51,76].

Participants: 15 patients and their carers/relatives and 15 practitioners/service planners (see 2.2).

Service planners including clinical leads, Integrated Care Boards (ICB), Primary Care Network (PCN), GP Cluster and Health Board, and Allied Health

Professional (AHP) representatives will be identified from urban and rural geographically diverse Integrated Care Boards, primary and community services and professional networks (e.g., Research Development Forum).

Methods: Participants will meet initially in separate online/in-person patients/carer and professional groups to show the composite film, present the findings from 2.1 and 2.2 and identify priorities/needs/ideas. Together, they will meet to agree priorities and generate the intervention resources and provider training content. They will explore potential implementation barriers and enablers. A preliminary programme theory will be developed and materials for patients and provider training finalised. The new co-designed intervention, provider training and materials will be specified using TIDieR[77]

2.4 Aim: To develop a funding application to evaluate the new supported self-management intervention

Based on 2.1-2.3, I will work with public advisors and other stakeholders to consider trial design (e.g., outcomes, data collection methods, comparison group). I will apply for funding to investigate effectiveness, cost-effectiveness and implementability of the intervention.

Output: (i) Co-designed self-management intervention for CLTI patients. (ii) trial funding application.

Workstream 3 Supported home-based walking exercise behaviour change programme for people with intermittent claudication (months 1-24)

This workstream is part of an **NIHR PGfAR application (stage 2)** and builds on the MOSAIC1 trial (n=190, 6 sites in Southeast England).[57]

Question: Can a home-based walking exercise behaviour change programme delivered by multidisciplinary healthcare and non-traditional healthcare exercise providers in primary care and community services improve walking capacity in IC patients?

3.1 Aim: to identify core determinants of long-term walking exercise in IC patients

Design: Secondary analysis of MOSAIC1 trial process variables[57]
Methods: Mediation analysis of MOSAIC1 trial psychological process variables[78,79] will identify determinants of long-term walking exercise informed by behaviour change[80] and implementation theories[70]

3.2 Aim: to Identify factors that influence walking exercise and implementation barriers and facilitators from patient and provider perspective.

Design: Framework analysis of interviews from 20 MOSAIC1 trial participants and 14 physiotherapists will identify critical factors influencing longer term walking exercise behaviour change and factors that influenced intervention delivery and barriers/enablers to implementation[56,57].

Using findings from 3.1 and 3.2, with the public advisory group the modifiable determinants will be prioritise using COM-B[80], refine the MOSAIC1 programme theory (figure 1) and identify core behaviour change techniques[81].

3.3 Aim: to adapt MOSAIC2 for delivery by multidisciplinary healthcare and non-traditional healthcare exercise providers in primary care and community services

Design: Participatory co-design approach
Participants: 15 patients/carers, including participants from MOSAIC1, and 15 practitioners working in primary and community services and service planners (see 2.2, 2.3).

Methods: Guided by the ADAPT framework to identify the key features of the context and population that require adaptation[82], participants will initially work in separate groups and then together to adapt MOSAIC1, consider potential implementation barriers and facilitators in primary and community care by a range of healthcare professionals and nontraditional healthcare exercise providers. MOSAIC2 intervention, provider training and materials will be described using TIDieR[77] and consider integration in to the PAD care pathway.

3.4 Aim: to assess the feasibility of conducting the MOSAIC2 trial in primary and community care by a range of health professionals and nontraditional healthcare exercise providers

The feasibility of recruitment and randomisation was tested in MOSAIC1 trial[57]. This study will therefore focus on examining fidelity, acceptability and implementation outcomes and determinants of MOSAIC2 and provider training in primary care and community services.

Design: Prospective single-arm study
Participants and setting: 32 diverse adults aged over 50 years with IC from 6 geographically diverse sites over 6-months.

Interventions: All participants will receive MOSAIC2. It is envisaged that MOSAIC2 will comprise up to six in-person, remote or telephone sessions (patients' preference) delivered over 3-months by a range of trained providers.
Assessment timepoints: Baseline and 3 months (post-intervention)
Outcomes: Fidelity of delivery, core components and proficiency of motivational interviewing, fidelity of receipt and enactment assessed in a random sample of audio-recorded sessions. Completeness of clinical outcome data collection including 6-minute walk distance and physical activity level using accelerometery. Acceptability & implementability assessed using mixed methods. Inclusive recruitment practices assessed by diversity of participant characteristics.

3.5 Aim: To develop a funding application to evaluate MOSAIC2
A funding application to evaluate MOSAIC2 inclusivity, effectiveness and cost-effectiveness will be developed with the public advisory group.

Outputs: (i) Adapted and feasibility tested walking exercise behaviour change intervention for IC patients delivered in primary and community care (ii) study funding application

---

Workstream 4 Co-designing an mhealth intervention to promote physical activity (months 7-42)
This workstream will be supported by a new collaboration with Dr Tomas Vetrovsky (Czech Republic).

Question: Can an inclusive mhealth intervention be co-designed to promote physical activity for PAD patients

4.1 *Aim: to explore preferred activity, sedentary behaviour patterns, potential barriers and facilitators of using m-health interventions in PAD patients/carers.*

Design: qualitative interviews
Participants: A purposive sample of up to 30 patients and their carers/relatives (see 2.2).

Methods: Online/in-person/telephone semi-structured interviews (participant preference) to explore preferred physical activity, patterns of sedentary behaviour, experiences and potential barriers and facilitators of using mhealth interventions including activity trackers. Common themes will be identified using reflexive thematic analysis and key behavioural targets identified using COM-B[75,80].

4.2 *Aim: to explore practitioners' experience of mhealth interventions in community settings, useful features and potential barriers and facilitators for implementation*

Participants: A purposive sample of up to 15 multidisciplinary practitioners (see 2.2).
Methods: Two online 1.5 hour focus groups (6-8 participants each) will explore practitioner experiences of using and integrating m-health interventions and activity trackers in community practice, useful/redundant features, potential implementation barriers and facilitators. Common themes will be identified using reflexive thematic analysis[75].

4.3 Aim: *To co-design an mhealth intervention, including a personalised message library digital literacy and accessibility*

Design: participatory co-design approach
Methods: Drawing on 2.3 and 4.1-4.2, 15 diverse patients/carers and 15 practitioners will work in small co-design groups and together to review key behavioural targets, intervention format, components, support and communication channels. A key focus will be on digital inclusivity and accessibility. A published library of PAD physical activity behaviour change messages will be reviewed and tailored (100-150 messages)[66]. A sub-group will identify key provider training components.

4.4 Aim: *To explore comprehension and perceived usefulness of messages*

Design: Cognitive 'think aloud' interviews
Participants: Up to 10-15 diverse PAD patients (see 2.2).
Methods: Using a cognitive interviewing approach, participants read a set of randomly selected messages and prompted to verbally express their thoughts, interpretations and perceived usefulness of the messages[66,69]. Responses will be analysed thematically using an adapted coding category[69]

4.5 *Aim: To adapt and integrate the prototype mHealth intervention into an existing digital platform*
Design: Software engineering and system integration
Methods: The prototype intervention will be implemented using an existing mHealth platform, e.g., HealthReact, developed by collaborators at the University of Hradec Kralove, Czechia. This platform will continuously collect and evaluate data from a linked validated consumer activity tracker (e.g., Garmin, Fitbit) and trigger tailored messages. Messages can be triggered either at specific times and places; by the patient's behaviour (e.g., after a prolonged

sitting or when the patient begins walking) or by the patient's daily or weekly activity level. The platform will allow configuration of triggering parameters, e.g. duration between messages, number messages/day. monitor, automatic syncing with a cloud service accessible via a public Application Programming Interface.

4.6 *Aim: To refine prototype mhealth intervention*
Design: Iterative refinement of mhealth intervention using plan-do-study-act cycles
Methods: Up to 15 diverse PAD patients will use the prototype intervention for one week. Quantitative (was the intervention used? How much? real-time feedback after messages) and qualitative feedback (accessibility? ease of use? Timing and usefulness of messages?) will be obtained. The intervention will be refined between user testing cycles. Iterative cycles will continue until a finalised prototype is achieved.

4.7 *Aim: To assess the acceptability, inclusivity and immediate effect of finalised m-health intervention*
Design: prospective single arm study with embedded micro-randomisation and ecological momentary assessment (real-time data collection) sub-study.
Participants and setting: up to 30 diverse adults with PAD and 12 practitioners over 4-months from 6 community sites (see 2.2).
Intervention: All participants will receive the 6-week mhealth intervention by a provider trained to deliver intervention and support inclusivity.
Outcomes: At baseline and 6 weeks including physical activity, Walking Impairment Questionnaire[83], Physical Activity Enjoyment Scale[84], walking self-efficacy scale[85,86], quality of life[87], and symptom severity (e.g., claudication pain scale[88]) and acceptability (acceptability of Intervention Measure, the Intervention Appropriateness Measure, and Feasibility of Intervention[89]). Interviews with 50% participants and practitioners will explore inclusivity, acceptability and implementability.

Sub study: Using micro-randomisation (50% messages randomly selected for triggering), (i) real-time surveys will explore immediate views on message relevance and usefulness (ii) short-term impact of messages on activity and perceived relevance and usefulness of messages following triggered versus non-triggered messages.

4.8 Aim: To develop a funding application to assess inclusivity, clinical and cost-effectiveness of the mhealth intervention

I will work with the public advisory group and collaborators to develop a funding application to evaluate inclusivity, effectiveness and cost effectiveness of the mhealth intervention alone or integrated into an existing exercise behaviour change intervention (e.g., MOSAIC2)

Output: (i) a prototype mhealth intervention PAD patients (ii) funding application

Workstream 5 Development of an implementation strategy (month 48-60)

Question: How can our understanding of the context, mechanisms and outcomes be integrated into an implementation toolkit that can be used to influence ongoing spread and sustainability PAD community interventions

Aim: to co-design preliminary implementation strategies and guidance pack for inclusive community-based PAD rehabilitation

Design: Participatory co-design approach

Methods: Participants: (i) up to 15 PAD patients and carers, (ii) up to 30 professionals including community providers, and local/national service planners such as primary car networks,
Drawing on interview and focus groups data (workstreams 2-4), key implementation determinants for successful, and sustainable delivery of each intervention will be summarised using the updated CFIR[70].
Two workshops will be conducted. In the first workshop, the groups will (i) review and discuss the implementation barriers and facilitators for each intervention (ii) prioritise the barriers and enablers considering their importance and feasibility in different settings for successful, sustainable delivery of PAD community intervention. In the second workshop, the participants will consider implementation strategies, generated by the ERIC-CFIR Implementation matching tool[90], and brainstorm additional implementation strategies to address prioritised barriers and enablers. Strategies will consider operationalisation, accessibility and inclusivity.

After the meetings, the implementation strategies will be collated, guided by Proctor's recommendations for specifying and reporting implementation strategies[91], and an Implementation and Evaluation Toolkit to support scale and spread of PAD community interventions will be developed.

Output: tailored implementation strategies and guidance pack for PAD community interventions

Timelines for delivery

6. Outputs, dissemination, anticipated impact.

Outputs

Policy and service recommendations

If effective, this research will transform PAD community rehabilitation by providing co-designed interventions, provider training and implementation strategies and guidance pack for service planners, guideline developers and local and national policy makers. Progress and findings will be succinctly summarised in brief reports and presentations.

Practitioners, professional societies, and translation to NHS/international care

This research will produce scalable community rehabilitation for adaptation for different contexts and populations. The free provider training and intervention resources will be shared online with community and vascular practitioners to support the translation into NHS. Presentations will be delivered to vascular, rehabilitation and community service practitioners nationally and regionally. Through collaborations, societies and networking e.g., World Rehabilitation Alliance, I will contextualise and raise the research visibility through presentations and in-person meetings to transform care internationally.

Public, voluntary organisations and communities

I will host webinars to mobilise findings with participants, patient and public organisations. Study summaries, infographics, and animations in multiple languages of our research progress and findings will be accessible on YouTube and social media. I will host community meetings and will share information via faith centres and community and voluntary organisations (e.g., the national vascular public advisory group) as these are important dissemination channels to underserved communities.

Academic outputs

Open-access publications and conference presentations/abstracts will target national/international vascular, primary and community care, public health, health services audiences. I will develop a study website, and online study repositories to collate outputs and publicise progress.

Engaging with patient, NHS, and wider population

Stakeholder mapping identified the people and communities with the highest influence over and interest in how this research progresses and findings are interpreted and implemented. With the public advisory group, co-ordinator, practitioners, research and mentorship team, I will co-design tailored

mobilisation and communication strategies for each stakeholder group. These approaches will be refined as the programme progresses, but it is envisaged that it will include both breadth and depth knowledge mobilisation approaches informed by appropriate theories and frameworks[92]. This will include updating progress and results to participants, public and patient organisations using creative methods such as public blogs, webinars, infographics and social media (e.g., PAD Facebook group). I will work directly with Health Innovation Networks, Applied Research Collaborations, and continuously update service planners at Integrated Care Boards and NHS England using briefings, via RD forum conference attendance and existing meetings.

Future funding and support

Funding for three PAD community rehabilitation interventions trials will be sought during this professorship. As care moves from secondary to community care, service planners will be focussed on scalable interventions, costs and outcomes. I am committed to ensuring that costs are minimal, resulting in long-term healthcare system savings. The intervention materials and training will be freely available and deliverable at scale. No income generation is planned. Low awareness or poor implementation may demand further funding to support implementation.

Possible barriers for further research, development, and implementation

I will lead studies which address questions of critical importance, meaning the findings are more likely to be implemented. Potential barriers to research and digital participation will be mitigated by a dedicated workstream (workstream1: research inclusion) and by collaborating with experts with a proven track record of developing inclusive mhealth interventions If effective, new community services would require commissioning within NHS care and appropriate workforce training; this will need to be resourced for rollout.

Impact of this research and for whom?

This research will improve the lives of substantial number of underserved PAD patients (e.g., people with CLTI, older adults encountering digital exclusion, some ethnic or socioeconomically deprived communities). It addresses unmet care needs, key evidence gaps and population and funder research priorities[29,44]. It will improve wellbeing, physical activity, prevent deaths, lower limb amputations and other cardiovascular complications. It aligns with the Department of Health and Social Care and new government's focus on community and primary healthcare delivery [45,47]. This research will shape national/international community rehabilitation practice and PAD guidance[29], NHS care and this clinical context internationally. By reducing the need for more

intensive treatments, emergency attendances and admissions, this research will generate significant NHS cost savings. This research will develop research capacity in underrepresented disciplines by developing postdoctoral fellows and PhD students that are likely to be nursing or allied health professionals. It will provide doctoral supervision experience, funding application writing experience and leadership development.

7. Project Management

I will lead and be responsible for overall research delivery on time and within budget. I will monitor progress and expenses against the milestones (see Gantt chart) and report to the funder. I will meet the research team (research fellows, research inclusion co-ordinator) weekly. The research inclusion co-ordinator will work with me to deliver workstream 1, first post-doctoral researcher will lead on workstream 2 and 5, the second post-doctoral researcher will lead on workstream 3 and, with collaborators, the doctoral student will lead on workstream 4. These researchers will complete protocol development, ethical approvals, research delivery, report writing. They will lead subsequent funding applications, set-up and delivery of funded trials, with one or more possibly moving to work on them full-time.

I will meet my host mentor weekly initially then monthly. My professorship steering committee will meet twice every year. I will work with Professor Busse and colleagues at the Cardiff University Centre for Trials Research, who will support trial design and funding application development. They will develop, test and maintain the study databases, and monitor conduct/provide support and advice in the set-up, delivery and closedown of funded studies. The research host and sponsor will be City St George's, University of London who will oversee the implementation of all aspects of the research and ensure all contractual, legal, financial and regulatory obligations are achieved.

8. Ethics

All research will be conducted in accordance with relevant ethical, regulatory and legal requirements, including Good Clinical Practice, UK Policy Framework for Health and Social Care Research and laws governing data protection. Appropriate research governance will be obtained from City St George's ethic board, NHS Research Ethics Committee/Health Research Authority (HRA) approval as required.

9. Success criteria:

- Successful recruitment to all research posts
- Diverse patient, practitioner involvement, participation and engagement
- Reaching workstream milestones (without significant delays)

- Building research capacity in underrepresented disciplines

Barriers:

- Research inclusion (likelihood: low; impact: high): I will employ a dedicated co-Ordinator, regularly monitor diversity and identify ways to maximise research inclusion.
- Staff recruitment (likelihood: low; impact: high: The posts are attractive because of the topic and duration of funding available. I will use my networks (including incubator hub) to promote the roles
- Research set-up and management (likelihood: low; impact: high): I have experience achieving favourable ethical opinions via City St George's, the IRAS and NIHR Coordinated System for gaining NHS Permissions systems.

Please describe how patients/service users, carers and the public have been involved in developing this proposal (353/350 words)

The public have framed this professorship so that research inclusion at its core. Regularly, over two years, I discussed my research with diverse PAD patients and their carers in coproduction meetings that have shape the focus, research questions, and plan. Nineteen people with lived experience of PAD, including people from African Caribbean, South Asian, and Eastern European communities and those with CLTI shared their experiences of living with PAD and their treatment. To reach people from underserved communities who did not want to join a group or those experiencing digital exclusion, I talked to them individually via telephone or in-person.

They discussed the barriers to receiving rehabilitation in large hospitals (e.g. time, travel) and how they wanted to receive rehabilitation/support locally that was co-ordinated and responsive. They described how this research was 'much needed' as this part of NHS care was 'unacceptable and burdensome' and they asked for 'better access' to rehabilitation and ways to help them look after their condition and symptoms independently.

Patients with CLTI described having treatment in hospital and then 'being left in a vacuum at home'. They welcomed sessions with a provider to help them 'make the most of their lives'. People with IC told me that home-based walking exercise was often 'overlooked as a treatment', but that specific walking advice with support was needed. They liked the content and format of MOSAIC1 and commented that having this intervention provided in the community with 'enough time with the provider' was key to helping them walk more. They didn't mind who treated them 'just as long as they were well trained and supervised'.

All advisors with lived experience of PAD recognised that becoming more active was important and wanted to explore the mhealth using activity trackers, although some were nervous about this being the only treatment option. They were worried about 'everything going online' and suggested developing an intervention that was easy to use and had support 'on the end of a phone'. This advice changed the focus of this workstream. My advisors helped identify all research questions and develop the research plan. Three advisors read the public summary.

Please describe the ways in which patients/service users, carers and the public will be actively involved the proposed research, including any training and support provided (357/350 word limit

My public involvement strategy aligns with the NIHR 2025-2030 strategic commitments and six UK Public Involvement Standards, ensuring valuable, sustainable relationships, quality and consistency[93,94].

Standards 1-3: Our advisory group includes men, women and members of underrepresented populations (n=6). I will expand our advisory group and seek further representation from underserved populations and diverse geographical locations by continuing to work with patient and community groups. I will employ a dedicated research inclusion officer to plan, support and liaise with our advisors and community groups, co-ordinate our regular meetings and ad hoc activities. I will be responsible for oversight, planning and reporting of involvement, impact, and diversity in involvement, participation and engagement. I will offer research training and cultural competency training for public advisors, as required.

Standards 4-6: I will meet with my public advisors every 6-months to discuss workstream progress, and work with them in between on focused activities. I aim to strike a balance between meaningful involvement, without over-burdening them. Digital exclusion will be mitigated by providing data packages and technical support, or if a member is not able to join the larger online meeting, they will still be sent materials and can speak to me individually either remotely or via telephone. Advisors will be offered the opportunity to be involved in elements of workstreams that particularly interest them and invited to monthly research team meetings. Planned activities include reviewing and providing feedback on public facing materials including websites, reviewing study processes to enhance acceptability and experience of participation, promoting public awareness of the study, reaching hidden communities and dissemination. I will encourage our public advisors to co-present at conferences and become co-investigators on funding applications.

Standard 6: Public involvement will be monitored, reviewed and reported at workstream management meetings, with public advisors providing feedback, reviewing and agreeing reported impact. This will enable reflection on our processes to improve advisor experience and explore quality and impact. We will report our public and patient involvement in research using the Guidance for Reporting Involvement of Patients and the Public 2- short-form checklist[73]. All involvement activity and expenses will be reimbursed in line with the NIHR payment policy.

Leadership and Development (887/1000)

In 2017, I completed the King's College London emerging leader programme and, in 2021, the St George's, future leaders programme, Tangible benefits of the programme have been greater national and international research collaborations and research leadership. My leadership development is continuing in: health service research, community rehabilitation service delivery, capacity development and management and leadership at an international level.

Using Vitae's Researcher Development Framework, I have developed a leadership development programme, focusing on:

- *Transition to senior strategic leadership and personal effectiveness: choosing commitments wisely (Domain B, C)*
- *Maximising international networks, engagement and impact (Domain D)*
- *Consolidating and expanding knowledge (Domain A).*
- *Sustainable research capacity development (Domain D)*

This professorship will be a career step change into international strategic research leadership aligned with the NIHR aims for quality rehabilitation research. It provides senior training, development opportunities to work with local and national policy makers, fostering exciting new collaborations through research visits and networking opportunities.

Over the next 5 years I will:

Amplify my senior strategic leadership and personal effectiveness: as an international rehabilitation research leader that leads and influences others to bring about change across boundaries, with the **NIHR Future Focused Leadership Programme** (in year 1).

To consolidate my strong and dynamic research leadership, I will be mentored by Professors Mallen (primary care), Prof Busse (clinical trials), Prof Modarai (vascular) and Prof Jones (inclusive rehabilitation research, co-design). My personal development will be further supported by quarterly remote/in-person **leadership coaching** with Dr Carol Daniels, a coach with extensive experience of developing female academic leaders, throughout the 5-year professorship. The sessions will build personal effectiveness and leadership capability in key areas e.g. nurturing and developing others; influencing, mobilising individuals and groups; prioritisation and self-management; delegation and execution. This will enable me to transition to a more senior role towards the end of the professorship and understand the leadership needs of others in my teams.

International engagement, influence and impact: Leading at the level of professor will require me to engage and influence many different individuals and groups nationally and internationally. I plan to visit two leading PAD researchers: Prof Mary McDermott (Northwestern University) and Prof Andrew Gardner (Penn State College) in the USA (year 2). These researchers lead groundbreaking inclusive community-based PAD studies and transformed clinical services by introducing targeted interventions for underserved groups. Coproducing research and learning from them will directly inform my inclusive PAD research programme and spark new international collaborations.

To reinforce my knowledge and ensure that I remain abreast of current developments in remote physical activity monitoring and mhealth, I will join the Faculty of Physical Education and Sport, Charles University, Prague, Czech Republic (in year 2). Dr Tomas Vetrosky, a new collaborator (workstream 3), is a leading European researcher in mHealth and physical activity trials in people with long-term conditions. Collaboration with his centre will include further learning on ecological momentary analysis, digital regulations and will spearhead a new European research collaboration.

To further my international research and leadership profile, and networks, I will attend the Society for Vascular Surgery, World Physiotherapy, World Congress on Rehabilitation, International Primary Care and Accelerating Health Equity annual Conferences.

Building knowledge and skills: To complement my existing clinical research skillset, I will work alongside Cardiff University Centre for Trials Research (mentor: Busse) to learn about innovative rehabilitation trial designs and identifying and developing inclusive, meaningful outcomes for community rehabilitation trials. I will also schedule the 5-day course in advanced clinical trial design (adaptive, platform, sustainable trials) at Birmingham University (Year 3).
In pioneering the shift of PAD rehabilitation into community services, I will participate in one-to-one coaching with Dr Catherine Huffington, Director of Health improvement Southwest London ICB to strengthen my knowledge of planning and commissioning community services.

Sustainable research capacity development: I will build a sustainable inclusive strategy for rehabilitation research capacity development within my Professorship that builds on my current and continuing researcher development work at City St George's and future proof the NIHR community rehabilitation incubator hub that I lead. I will develop the next generation of

postdoctoral and doctoral scientists to become future community rehabilitation research leaders, building capacity through their own successful personal fellowship and funding applications.

Career intentions

This Professorship will give me an extraordinary opportunity to deliver exceptional and much-needed community rehabilitation research to transform the care of PAD patients. City St George's has committed to continuing to support me and my research beyond this award through access to training opportunities for current and new researchers in my group and providing administrative and infrastructural support. At the end of this award, my goals are to

- be leading a multidisciplinary, collaborative team who will maximise inclusive research and support the delivery and mobilisation of my international, high impact research
- have developed inclusive PAD community interventions to transform community services and built on through securing further programme grant funding
- move from national lead in research and knowledge mobilisation in practice to enabling high quality methodological research to be done at a pace
- increase my national and international profile, enabling me to maximise the opportunities to contribute to international guideline groups, grant panels (e.g. HTA), and to speak at high-profile meetings where impact is greatest
- have established a sustainable collaborative strategy and network to nurture the future generations of community researchers at all career stages
- be a strong candidate for NIHR Senior Investigator status

Mentors

Professor Fiona Jones (host mentor)

Professor Christian Mallen

Christian Mallen is a past NIHR Research Professor, NIHR Senior Investigator and Director of the NIHR School for Primary Care Research. He is an academic general practitioner with a portfolio of research focusing on improving the diagnosis and management of musculoskeletal and rheumatological disorders in community settings. He has a strong track record in multi-disciplinary working and supporting career development, as past academic capacity development lead for the NIHR SPCR and director of the Wellcome PhD

Programme for Primary Care. His research group is multi-professional including clinicians (physiotherapist, nurses, pharmacists, psychologists and GPs) and methodological experts (clinical trials, epidemiology, data synthesis and social sciences).

Professor Mallen's mentorship will fulfil three essential roles in my future career development.
Firstly, as a leading international expert in primary and community care, where my research is focused, he will ensure my research remains relevant to the needs of patients and primary and community care practitioners and is developed to be applicable in that setting.

Secondly, his experiences of developing an internationally renowned multidisciplinary programme of research and his leadership role with the NIHR SPCR will provide a unique perspective on academic training and capacity building which will support me to build a strong cohort of multi-disciplinary researchers.

Thirdly, he will provide opportunities for formal collaboration with world-leading general practice researchers, for example, Professor Paul Little, University of Southampton

Professor Mallen will join the professorship steering committee that meets twice every year and be available for ad hoc advice via remote meetings or email. **249/250 words.**

Professor Monica Busse
Professor Monica Busse is an applied health researcher and trial methodologist based in the Cardiff University Centre for Trials Research, Director of the Health and Care Research Wales Faculty, and current Health and Care Research Wales Senior Research Leader (Senior Investigator equivalent). Her research seeks to enable people living with chronic health conditions to sustain an active lifestyle while staying connected and engaged in meaningful aspects of their lives. In CTR, she is Chief Investigator (or co-CI) of a suite of innovative NIHR and charity/government funded rehabilitation trials. Prof Busse is an accredited executive coach with an interest in promoting research careers and developing future research leaders and is ideally placed to mentor me as I move through this professorship and programme of work.

Specifically, I will draw on her experience, knowledge and insight of developing an internationally renowned multidisciplinary research programme when

- designing and delivering innovative trials. Prof Busse has agreed to host a research visit to the Centre for Trials Research
- developing strategy and building a strong culture of inclusion and a positive working environment.
- planning and enabling research capacity development of my team and others in my broader network

Together with Professor Jones, she will act as a role model and nurture my development as a strong female research leader. Prof Busse is a vital member of my professorship steering committee that will meet twice every year. Prof Busse will also be available for ad hoc advice via remote meetings or email

Professor Bijan Modarai

Professor Bijan Modarai is the King's Chair in Vascular Surgery, a past British Heart Foundation Senior Clinical Research Fellow, current President of the British Society for Endovascular Therapy and leads the King's Health Partners Cardiovascular Clinical Academic grouping. He is an academic vascular surgeon whose work focusses on improving diagnosis and management of people with vascular disease by developing novel diagnostics and advanced therapeutics. He is an advocate of "team science", leading translational programmes of vascular research that engages multi-disciplinary collaborative teams, including clinicians (vascular scientists, physiotherapists, nurses) and vascular biologists, data scientists, statisticians, imaging experts and radiation biologists. He is an internationally recognised clinical-academic with an extensive network of international collaborations that he has leveraged for impactful outputs. I will build on our existing relationship, cultivated from collaborating with him on several research projects since 2010. I am honoured that he has agreed to provide mentorship through my transition to professorship. Specifically, his mentorship will focus on:

- my engagement with international collaborators to highlight areas of unmet need and establish vascular research
- emerging areas and the wider context of management of vascular patients
- through his extensive international collaborative network, he will connect me with vascular surgeons who lead highly impactful research and broker new research collaborations e.g., Kevin Mani Professor of Vascular Surgery, Uppsala, Sweden

He will contribute to my professorship steering committee that meets twice every year and be available for ad hoc remote advice and direction on both the science and my career trajectory. **(248/250 words)**

Collaborations (525/600)

My Professorship will extend established and successful partnerships and create important new collaborations.

I will forge new *international collaborations* in:

United States of America: Prof Mary MacRae McDermott, Northwestern University's Feinberg School of Medicine and Northwestern Memorial Hospital, Chicago. Prof McDermott and her team have undertaken the flagship HONOR and GOALs trials into community supported home-based exercise/mhealth in PAD. Prof Andrew Gardner, Department of Physical Medicine and Rehabilitation at Penn State College of Medicine, is an international researcher who led seminal trials comparing revascularisation and exercise effects in PAD. His current work investigates PAD sub-populations such as people with PAD/diabetes and ethnic communities. By working alongside these teams, I will be able to gain an in-depth understanding of the interventions, questionnaires and data used current and other studies, which will inform workstream 1-2, support work coming out of workstream 3-4, and will spark/lead onto other avenues for international collaboration.

The Czech Republic: Dr Tomas Vetrovsky, Charles University, Prague is an expert in physical activity behaviour change using inclusive digital interventions. His knowledge from WATCHFUL and ENERGISE trials that use consumer activity trackers with behaviour change prompts will support workstream 4. We plan to pool experience from our previous trials to maximise research inclusion and participation,

I will build on existing collaborations with clinicians, methodologists and patients in the vascular and rehabilitation communities: I will continue to link with the Professor Janet Peacock, Professor of Epidemiology and Biomedical Data Science and Professor Jeremiah Brown, Centre for Implementation Science at Dartmouth Geisel School of Medicine, USA to ensure my interventions and strategies that are robust, effective, feasible, and sustainable in real-world settings.

In the UK, I will continue to work closely with the following clinicians and researchers: Dr Amy Harwood, Reader in clinical exercise physiologist, Manchester Metropolitan University; Prof Athanasios Saratzis, Professor of Vascular Surgery and Prof Charlotte Edwardson, Professor of Physical activity, sedentary behaviour and health, University of Leicester; Dr Louise Hull, implementation Scientist at King’s College London, co-investigators on my MOSAIC2 NIHR PGfAR application. I will maintain my strong working

relationship with clinical and methodological colleagues at City St George's, University of London: Professor Michael Ussher, Dr Mandeep Sekhon, health psychologists/behaviour change experts, Professor Tess Harris, GP and Professor of Primary Care Research and primary care physical activity triallist

Partnering with research inclusion experts: Having established relationships with UK patient organisations for vascular conditions to maximise research inclusion (e.g., the national vascular public advisory group), I would like to consolidate my embryonic relationships with communities that work with typically underserved groups: the Centre for Ethnic health research and the Romanian and Eastern European Hub. I will extend my links to other organisations such as Caribbean & African Health Network, an organisation which seeks to address long-standing health inequalities disadvantaging people of Black Caribbean and African heritage.

I will build on my collaborations with local and national service planners by continuing to work with the Southwest London Integrated Care Board research group and developing specific relationship with Dr Catherine Heffernan, Director of Health Improvement and her primary and community services commissioning team. This will ensure that all my research is relevant to local and national community-based needs, is user driven and fit for purpose

References (To be converted to 1 page short references)

1. Mazzolai L, Belch J, Venermo M, et al. Exercise therapy for chronic symptomatic peripheral artery disease: A clinical consensus document of the European Society of Cardiology Working Group on Aorta and Peripheral Vascular Diseases in collaboration with the European Society of Vascular Medicine and the European Society for Vascular Surgery. *European Heart Journal*. 2024;45(15):1303-1321. doi:10.1093/eurheartj/ehad734
2. Carroll SP, Gray SR, Wood L, Bearne L, Seenan C. Motivating Physical Activity with a Walking Exercise Behaviour Intervention and Pain Management Remotely in Intermittent Claudication (MAvERIC): protocol for a randomised controlled feasibility trial. *Journal of Vascular Societies Great Britain & Ireland*. 2024;3(3):147-154. doi:10.54522/jvsgbi.2024.128
3. Hendry GJ, Bearne L, M, Foster NE, et al. Gait rehabilitation for foot and ankle impairments in early rheumatoid arthritis: a feasibility study of a new gait rehabilitation programme (GREAT Strides). *Pilot and Feasibility Studies*. 2022;8(115)doi:10.1186/s40814-022-01061-9
4. Sekhon M, Thurah DA, Fragoulis EG, et al. Synthesis of guidance available for assessing methodological quality and grading of evidence from qualitative research to inform clinical recommendations: a systematic literature review. *RMD Open*. 2024;10(2):e004032. doi:10.1136/rmdopen-2023-004032
5. Bearne LM, Sekhon M, Grainger R, et al. Smartphone Apps Targeting Physical Activity in People With Rheumatoid Arthritis: Systematic Quality Appraisal and Content Analysis. *JMIR Mhealth Uhealth*. 2020;8(7):e18495. doi:10.2196/18495
6. Abaraogu UO, Dall P, Seenan C, et al. Effect of Behavior-change Interventions on Daily Physical Activity in Patients with Intermittent Claudication: The OPTIMA Systematic Review with

Meta-Analysis. *European Journal of Preventive Cardiology*. 2024:zwae296. doi:10.1093/eurjpc/zwae296
7. Stisen BM, Mechlenburg I, Bearne ML, Godfrey E, Pedersen BA, Sørensen D. Exploring needs, barriers to, and facilitators of rehabilitation exercise following revision hip replacement – A grounded theory study. *Disability and Rehabilitation*. 2024;46(21):5012-5020. doi:10.1080/09638288.2023.2295008
8. Gohil NM, Muruganantham B, Raval M, Bearne ML. Translation, cross-cultural adaptation, and validation of a Gujarati version of a theory of planned behavior questionnaire that assesses walking treatment beliefs in people with intermittent claudication. *Journal of Vascular Nursing*. 2024;42(3):182-190. doi:10.1016/j.jvn.2024.05.004
9. O'Sullivan JD, Bearne ML, Harrington MJ, Cardoso RJ, Mcveigh GJ. The effectiveness of social prescribing in the management of long-term conditions in community-based adults: A systematic review and meta-analysis. *Clinical Rehabilitation*. 2024;38(10):1306-1320. doi:10.1177/02692155241258903
10. Gimigliano F, Negrini S. The World Health Organization "Rehabilitation 2030: a call for action. *European Journal of Physical Rehabilitation and Medicine*. 2017;2017 (2)(53):155-168. doi:10.23736/S1973-9087.17.04746-3
11. Community Rehabilitation Alliance. *Making the right to rehabilitation a reality: Community Rehabilitation Alliance position statement*. 2021. cra_position_statement_2020-final_-_two_page_version.pdf
12. Lüscher TF. Peripheral arterial disease guidelines: the Cinderella of cardiovascular medicine gets the attention it deserves. *European Heart Journal*. 2018;39(9):717-719. doi:10.1093/eurheartj/ehy086
13. Song P, Rudan D, Zhu Y, et al. Global, regional, and national prevalence and risk factors for peripheral artery disease in 2015: an updated systematic review and analysis. *The Lancet Global Health*. 2019;7(8):e1020-e1030. doi:10.1016/S2214-109X(19)30255-4
14. Cea-Soriano L, Fowkes FGR, Johansson S, Johansson S, Allum AM, García Rodriguez LA. Time trends in peripheral artery disease incidence, prevalence and secondary preventive therapy: a cohort study in The Health Improvement Network in the UK. *BMJ Open*. 2018;8(e018184)doi:10.1136/ bmjopen-2017-018184
15. Dua A, Lee CJ. Epidemiology of Peripheral Arterial Disease and Critical Limb Ischemia. *Techniques in Vascular and Interventional Radiology*. 2016;19(2):91-95.
16. Morley RL, Sharma A, Horsch AD, Hinchliffe RJ. Peripheral artery disease. *British Medical Journal*. 2018;360: j5842.
17. Ali R, Chowdhury A, Forouhi N, N W. *Ethnic disparities in the major causes of mortality and their risk factors - a rapid review*. 2021. Ethnic disparities in the major causes of mortality and their risk factors – a rapid review - GOV.UK (www.gov.uk)
18. Sanna G, Alesso D, Meddiati M, et al. Prevalence of peripheral arterial disease in subjects with moderate cardiovascular risk: Italian results from the PANDORA study Data from PANDORA (Prevalence of peripheral Arterial disease in subjects with moderate CVD risk, with No overt vascular Diseases nor Diabetes mellitus). *BMC Cardiovascular Disorders*. 2011;11:1-10.
19. Gupta R, Misra A. Epidemiology of microvascular complications of diabetes in South Asians and comparison with other ethnicities. *Journal of diabetes*. 2016;8(4):470–482. doi:10.1111/1753-0407.12378
20. Parvar SL, Fitridge R, Dawson J, Nicholls SJ. Medical and lifestyle management of peripheral arterial disease. *Journal of Vascular Surgery*. 2018;68(5):1595-606.
21. McDermott MM, Greenland P, Liu K, et al. Leg symptoms in peripheral arterial disease: associated clinical characteristics and functional impairment. *JAMA*. 2001;286(13):1599-1606.

22. Cheema KM, Dicks E, Pearson J, Samani NJ. Long-term trends in the epidemiology of cardiovascular diseases in the UK: insights from the British Heart Foundation statistical compendium. *Cardiovascular Research*. 2022;118(10):2267 - 80.
23. Collins B, Bandosz P, Guzman-Castillo M, et al. What will the cardiovascular disease slowdown cost? Modelling the impact of CVD trends on dementia, disability, and economic costs in England and Wales from 2020-2029. *PLoS One*. 2022;17(6):e026876.
24. Steffen MW, Undavalli C, Asi N, et al. The natural history of untreated severe or critical limb ischemia. *Journal of vascular surgery*. 2015;62(6):1642-1651.
25. Dhaliwal G, Mukherjee D. Peripheral arterial disease: Epidemiology, natural history, diagnosis and treatment. *International Journal of Angiology*. 2007;16(2):36-44.
26. Nehler MR, Duval S, Diao L, et al. Epidemiology of peripheral arterial disease and critical limb ischemia in an insured national population. *Journal of Vascular Surgery*. 2014;60(3):686-95 e2.
27. Perlander A, Broeren M, Österberg K, Svensson M, Nordanstig J. Disease Specific Health Related Quality of Life in Patients With Chronic Limb Threatening Ischaemia Undergoing Revascularisation of Femoropopliteal Lesions. *European Journal of Vascular and Endovascular Surgery*. 2023/08/01/ 2023;66(2):245-251. doi:https://doi.org/10.1016/j.ejvs.2023.05.014
28. Pymer S, Harwood AE, Long J, et al. Research priorities for patients with peripheral arterial disease: a James Lind Alliance Priority Setting Partnership. *Journal of Vascular Societies of Great Britain and Ireland*. 2022;1(2):23-29. doi:10.54522/jvsgbi.2022.011
29. National Clinical Guideline Centre. Lower Limb Peripheral Arterial Disease - diagnosis and management (CG 147). Accessed 21/12/2021, https://www.nice.org.uk/guidance/cg147
30. Grewal P, Davis M, Hamilton G. Provision of vascular surgery in England in 2012. *European Journal of Vascular and Endovascular Surgery*. 2013/01/01/ 2013;45(1):65-75. doi:https://doi.org/10.1016/j.ejvs.2012.10.017
31. Horrocks M. *Vascular Surgery. GIRFT Programme National Specialty Report*. 2018.
32. World Health Organisation: Sensory Functions DaRt. *Access to rehabilitation in primary health care: an ongoing challenge*. 2019:27.
33. Bearne LM, Delaney N, Nielsen M, Sheehan KJ. Inequity in exercise-based interventions for adults with intermittent claudication due to peripheral arterial disease: a systematic review. *Disability and Rehabilitation* 2022:1-10. doi:10.1080/09638288.2022.2102255
34. Attwood S, van Sluijs E, Sutton. S. Exploring equity in primary-care-based physical activity interventions using PROGRESS-Plus: a systematic review and evidence synthesis. *International Journal of Behavioural Nutrition and Physical Activity*. 2016;13:60.
35. Abaraogu UO, Ezenwankwo EF, Dall PM, Seenan CA. Living a burdensome and demanding life: A qualitative systematic review of the patients experiences of peripheral arterial disease. *Plos One*. 2018;13(11):e0207456. doi:10.1371/journal.pone.0207456
36. Abaraogu U, Ezenwankwo E, Dall P, et al. Barriers and enablers to walking in individuals with intermittent claudication: A systematic review to conceptualize a relevant and patient-centered program. *PloS one*. 2018;13(7):e0201095.
37. Makris G, Lattimer C, Lavida A, Geroulakos G. Availability of supervised exercise programs and the role of structured home-based exercise in peripheral arterial disease. *European Journal of Vascular and Endovascular Surgery*. 2012;44(6):569-575. doi:10.1016/j.ejvs.2012.09.009
38. Kumar BN, Gambhir RP. Critical limb ischemia-need to look beyond limb salvage. *Ann Vasc Surg*. Oct 2011;25(7):873-7. doi:10.1016/j.avsg.2011.05.020
39. Jenkins N, Jhundoo N, Rainbow P, Sheehan KJ, Bearne LM. Inequity in exercise-based interventions for adults with rheumatoid arthritis: a systematic review. *Rheumatology Advances in Practice*. 2023;24(1)(7)doi:10.1093/rap/rkac095

40. Harwood AE, Pymer S, Ibeggazene S, Ingle L, Caldow E, Birkett ST. Provision of exercise services in patients with peripheral artery disease in the United Kingdom. *Vascular Medicine*. 2021:17085381211035259. doi:10.1177/17085381211035259
41. Elfghi M, Dunne D, Jones J, et al. Mobile health technologies to improve walking distance in people with intermittent claudication. *Cochrane Database of Systematic Reviews*. 2024; (2)doi:10.1002/14651858.CD014717.pub2
42. Lu X, Yao Y, Jin Y. Digital exclusion and functional dependence in older people: findings from five longitudinal cohort studies. *EClinicalMedicine*. 2022;4doi:10.1016/j.eclinm.2022.101708
43. Barker I, Steventon A, Williamson R, Deeny SR. Self-management capability in patients with long-term conditions is associated with reduced healthcare utilisation across a whole health economy: cross-sectional analysis of electronic health records. *BMJ Quality & Safety*. 2018;27(12):989. doi:10.1136/bmjqs-2017-007635
44. Best Research for Best Health: The Next Chapter. Our Operational Priorities, (UK Government) (2022).
45. DHSCs areas of research interest (UK Government) (2023).
46. A plan for digital health and care research (UK Government) (2022).
47. Darzi A. *Independent investigation of the National Health Service*. 2024:1-59.
48. Reynolds R, Dennis S, Hasan I, et al. A systematic review of chronic disease management interventions in primary care. . *BMC family practice*. 2018 19(1-3):1-3.
49. Dineen-Griffin S, Garcia-Cardenas V, Williams K, Benrimoj SI. Helping patients help themselves: a systematic review of self-management support strategies in primary health care practice. *PLoSOne*. 2019;14:e0220116.
50. Fu V, Weatherall M, McPherson K, et al. Taking charge after stroke: a randomized controlled trial of a person-centered, self-directed rehabilitation intervention. *International Journal of Stroke*. 2020;15(9):954-64.
51. Kulnik ST, Poestges H, Brimicombe L, Hammond J, Jones F. Implementing an interprofessional model of self-management support across a community workforce: a mixed-methods evaluation study. *Journal of interprofessional care*. 2017;31(1):75-84.
52. Kim SY, Lee YM, Son YJ. Factors associated with self-management after hybrid revascularization in patients with peripheral artery disease: A structural equations model. *Journal of advanced nursing*. 2023;79(1):170-81.
53. Thangada ND, Zhang D, Tian L, et al. Home-Based Walking Exercise and Supervised Treadmill Exercise in Patients With Peripheral Artery Disease: An Individual Participant Data Meta-Analysis. *JAMA Network Open*. 2023;6(9):e2334590-e2334590. doi:10.1001/jamanetworkopen.2023.34590
54. Pymer SA, Ibeggazene S, Palmer J, et al. An Updated Systematic Review and Meta-Analysis of Home-based Exercise Programmes for Individuals with Intermittent Claudication. *Journal of Vascular Surgery* 2021;74(6):2076-2085. doi:10.1016/j.jvs.2021.03.063
55. Twomey A, Khan Z. Home-Based Exercise Therapy in the Management of Intermittent Claudication: A Systematic Review and Meta-Analysis. *Cureus*. 2023;15(5)
56. Volkmer B, Sekhon M, Bieles J, et al. Participants' experiences and acceptability of a home-based walking exercise behaviour-change intervention (MOtivating Structure walking Activity in people with Intermittent Claudication (MOSAIC)). *Physiotherapy*. 2024/03/01/ 2024;122:70-79. doi:https://doi.org/10.1016/j.physio.2023.09.002
57. Bearne LM, Volkmer B, Peacock J. Effect of a Home-Based, Walking Exercise Behavior Change Intervention vs Usual Care on Walking in Adults With Peripheral Artery Disease: The MOSAIC Randomized Clinical Trial. *Journal of American Medical Association*. 2022;327(14):**1344-1355**. doi:10.1001/jama.2022.3391
58. Bearne L, Galea Holmes M, Bieles J, et al. Motivating Structured walking Activity in people with Intermittent Claudication (MOSAIC): protocol for a randomised controlled trial of a

physiotherapist-led, behavioural change intervention versus usual care in adults with intermittent claudication. *BMJ open*. 2019;9(8)doi:10.1136/bmjopen-2019-030002
59. Galea Holmes MN, Weinman JA, Bearne LM. A randomized controlled feasibility trial of a home-based walking behavior–change intervention for people with intermittent claudication. *Journal of Vascular Nursing*. 2019;37(2):135-143. doi:10.1016/j.jvn.2018.11.001
60. World Health Organisation. *WHO guidelines on physical activity and sedentary behaviour*. 2020. 9789240015128.
61. Gerage AM, Correia M, Oliveira P, et al. Physical Activity Levels in Peripheral Artery Disease Patients. *Arquivos Brasileiros de Cardiologia*. 2019;113
62. Lauret GJ, Fokkenrood HJP, Bendermacher BL, Scheltinga MRM, Teijink JAW. Physical Activity Monitoring in Patients with Intermittent Claudication. *European Journal of Vascular and Endovascular Surgery*. 2014/06/01/ 2014;47(6):656-663. doi:https://doi.org/10.1016/j.ejvs.2014.03.001
63. Ferguson T, Olds T, Curtis R, et al. Effectiveness of wearable activity trackers to increase physical activity and improve health: a systematic review of systematic reviews and meta-analyses. *The Lancet Digital Health*. 2022;4(8):e615-e626. doi:10.1016/S2589-7500(22)00111-X
64. Novak J, Jurkova K, Lojkaskova A, et al. Participatory development of an mHealth intervention delivered in general practice to increase physical activity and reduce sedentary behaviour of patients with prediabetes and type 2 diabetes (ENERGISED). *BMC Public Health*. 2024/03/31 2024;24(1):927. doi:10.1186/s12889-024-18384-2
65. Vetrovsky T, Borowiec A, Juřík R, et al. Do physical activity interventions combining self-monitoring with other components provide an additional benefit compared with self-monitoring alone? A systematic review and meta-analysis. *Br J Sports Med*. Dec 2022;56(23):1366-1374. doi:10.1136/bjsports-2021-105198
66. Kim M, Kim Y, Choi M. Development of an Ecological Momentary Assessment-Triggered Text Message Library for Peripheral Artery Disease. *West J Nurs Res*. May 2023;45(5):469-477. doi:10.1177/01939459221150391
67. Skivington K, Matthews L, Simpson SA, et al. A new framework for developing and evaluating complex interventions: update of Medical Research Council guidance. *British Medical Journal*. 2021;374:n2061.
68. Goodwin VA, Low MSA, Quinn TJ, et al. Including older people in health and social care research: best practice recommendations based on the INCLUDE framework. *Age and Ageing*. 2023;52(6):afad082. doi:https://doi.org/10.1093/ageing/afad082
69. Treweek S, Banister K, Bower P, et al. Developing the INCLUDE Ethnicity Framework—a tool to help trialists design trials that better reflect the communities they serve. *Trials*. 2021/05/10 2021;22(1):337. doi:10.1186/s13063-021-05276-8
70. Damschroder LJ, Reardon CM, Widerquist MAO. The updated Consolidated Framework for Implementation Research based on user feedback. *Implementation Science*. 2022;17(75)doi:https://doi.org/10.1186/s13012-022-01245-0
71. Whittaker R, Merry S, Dorey E, Maddison R. A development and evaluation process for mHealth interventions: examples from New Zealand. *J Health Commun*. 2012;17 Suppl 1:11-21. doi:10.1080/10810730.2011.649103
72. Centre for Ethnic Health Research. *Equality Impact Assessment (EqIA) toolkit*. 2021.
73. Staniszewska S, Brett J, Simera I, et al. GRIPP2 reporting checklists: tools to improve reporting of patient and public involvement in research. *BMJ*. Aug 2 2017;358:j3453. doi:10.1136/bmj.j3453
74. Francis JJ, O'Connor D, Curran J. Theories of behaviour change synthesised into a set of theoretical groupings: introducing a thematic series on the theoretical domains framework. *Implementation Science*. 2012;7:35.
75. Braun V, Clark V. *Thematic Analysis - a practical guide*. SAGE Publications Ltd; 2021.

76. Jones F, Pöstges H, Brimicombe L. Building Bridges between healthcare professionals, patients and families: A coproduced and integrated approach to self-management support in stroke. *NeuroRehabilitation*. 2016;39:471-480.
77. Hoffmann TC, Glasziou PP, Boutron I, et al. Better reporting of interventions: template for intervention description and replication (TIDieR) checklist and guide. *The British Medical Journal*. 2014-03-07 2014;348:g1687. doi:10.1136/bmj.g1687
78. Darker CD, French DP. What sense do people make of a theory of planned behaviour questionnaire?: A think-aloud study. *Journal of Health Psychology*. 2009;14(7):861-871. doi:10.1177/1359105309340983
79. Broadbent E, Petrie KJ, Main J, Weinman JA. The brief illness perception questionnaire. *Journal of Psychosomatic Research*. 2006;60(6):631-637. doi:10.1016/j.jpsychores.2005.10.020
80. Michie S, Van Stralen MM, West R. The behaviour change wheel: a new method for characterising and designing behaviour change interventions. *Implementation science*. 2011;6(1):42.
81. Michie S, Wood CE, Johnston M, Abraham C, Francis JJ, Hardeman W. Behaviour change techniques: the development and evaluation of a taxonomic method for reporting and describing behaviour change interventions (a suite of five studies involving consensus methods, randomised controlled trials and analysis of qualitative data). *Health Technology Assessment*. 2015;19(99):1-188. doi:10.3310/hta19990
82. Moore G, Campbell M, Copeland L, et al. Adapting interventions to new contexts—the ADAPT guidance. *BMJ*. 2021:n1679. doi:10.1136/bmj.n1679
83. McDermott MM, Liu KA, Guralnik JM, Martin GJ, Criqui MH, Greenland P. Measurement of walking endurance and walking velocity with questionnaire: Validation of the walking impairment questionnaire in men and women with peripheral arterial disease. *Journal of Vascular Surgery*. Dec 1998;28(6):1072-1081.
84. Mullen SP, Olson EA, Phillips SM, et al. Measuring enjoyment of physical activity in older adults: invariance of the physical activity enjoyment scale (paces) across groups and time. *International Journal of Behavioral Nutrition and Physical Activity*. 2011/09/27 2011;8(1):103. doi:10.1186/1479-5868-8-103
85. McAuley E, Lox C, Duncan TE. Long-term maintenance of exercise, self-efficacy, and physiological change in older adults. *Journal of Gerontology (Psychological Science)*. 1993;48:218-224.
86. McAuley E, Courneya KS, Lettunich J. Effects of acute and long-term exercise on self-efficacy responses in sedentary, middle-aged males and females. *Gerontologist*. Aug 1991;31(4):534-42. doi:10.1093/geront/31.4.534
87. Nordanstig J, Wann-Hansson C, Karlsson J. Vascular quality of life Questionnaire-6 facilitates health-related quality of life assessment in peripheral arterial disease. *Journal of Vascular Surgery*. 2014;59:700-707. doi:10.1016/j.jvs.2013.08.099
88. Parmenter BJ, Dieberg G, Smart NA. Exercise Training for Management of Peripheral Arterial Disease: A Systematic Review and Meta-Analysis. *Sports Medicine*. 2015/02/01 2015;45(2):231-244. doi:10.1007/s40279-014-0261-z
89. Weiner BJ, Lewis C, C, Stanick C, et al. Psychometric assessment of three newly developed implementation outcome measures. *Implementation Science*. 2017/08/29 2017;12(1):108. doi:10.1186/s13012-017-0635-3
90. McHugh S, Presseau J, Luecking CT, Powell BJ. Examining the complementarity between the ERIC compilation of implementation strategies and the behaviour change technique taxonomy: a qualitative analysis. *Implementation Science*. 2022/08/19 2022;17(1):56. doi:10.1186/s13012-022-01227-2
91. Proctor EK, Powell BJ, McMillen JC. Implementation strategies: recommendations for specifying and reporting. *Implementation Science*. 2013;8(139)

92. Nilsen P. Making sense of implementation theories. models and frameworks. *Implementation Science*. 2015;10(53)doi:10.1186/s13012-015-0242-0
93. National Insititute for Health and Care Research. *NIHR 2025-2030 strategic commitments* 2024.
94. The UK Public Involvement Standards Development Partnership. *UK Standards for Public Involvement*. 2019.